Targeted stimulus screen for model-specific focal points

Step 1c · GPT-5.4 vs Claude Opus 4.8 · one-shot visual coordination

2026-08-07 · 3,660 API calls · 27 new stimulus pairs screened, 17 confirmed

Question

Extend the coordination battery with stimuli on which GPT and Claude genuinely differ in focal strength — both models coordinating on FORM, one simply more concentrated — rather than stimuli where the weaker model has no form rule and quietly defaults to a side.

Headline findings

— Only two stimuli meet all five criteria on fresh data, and both run GPT > Claude: `d_dens_extreme`, `s_density`. There is no clean Claude > GPT disparity in the battery.

— The asymmetry is the main result. Every GPT > Claude gap is clean — Claude keeps form consistency 0.80–0.97 and merely concentrates less. Every Claude > GPT gap but one has GPT at form consistency 0.43–0.70: GPT failing to find a rule and choosing a side. These are differences in KIND, not in focal strength.

— The winner's curse bit. The screen's only clean disparity, b_spiral_cw, fell from Δ −0.303 to −0.176 on fresh trials, below the 0.20 bar; d_eq_ink collapsed from −0.303 to −0.048. At n=30 alone both would have entered the battery.

— Canonical orientation is a shared focal point, not a disparity. Six of seven pairs in family A sit at 1.000/1.000 for both models. s_arrow has no sibling that reproduces it, so it is an idiosyncratic GPT failure, not an orientation deficit.

— Three stimuli produce large, perfectly replicating Δ that are pure position defaults (b_tee_ell, c_contr_subtle, d_texture; weak-model form consistency 0.07–0.27). They are retained as diagnostics and never counted as focality disparities.

Procedure

Prompt, model settings and parser are imported verbatim from step1_form_choice.py (prompt sha8 3003ae1b), the module Step 1 and Step 1b used, so all three datasets are comparable. GPT-5.4 reasoning.effort=low; Claude Opus 4.8 max_tokens=2000, adaptive thinking, output_config.effort=low. Neither API exposes a sampling seed; fixed seeds govern AB/BA allocation only.

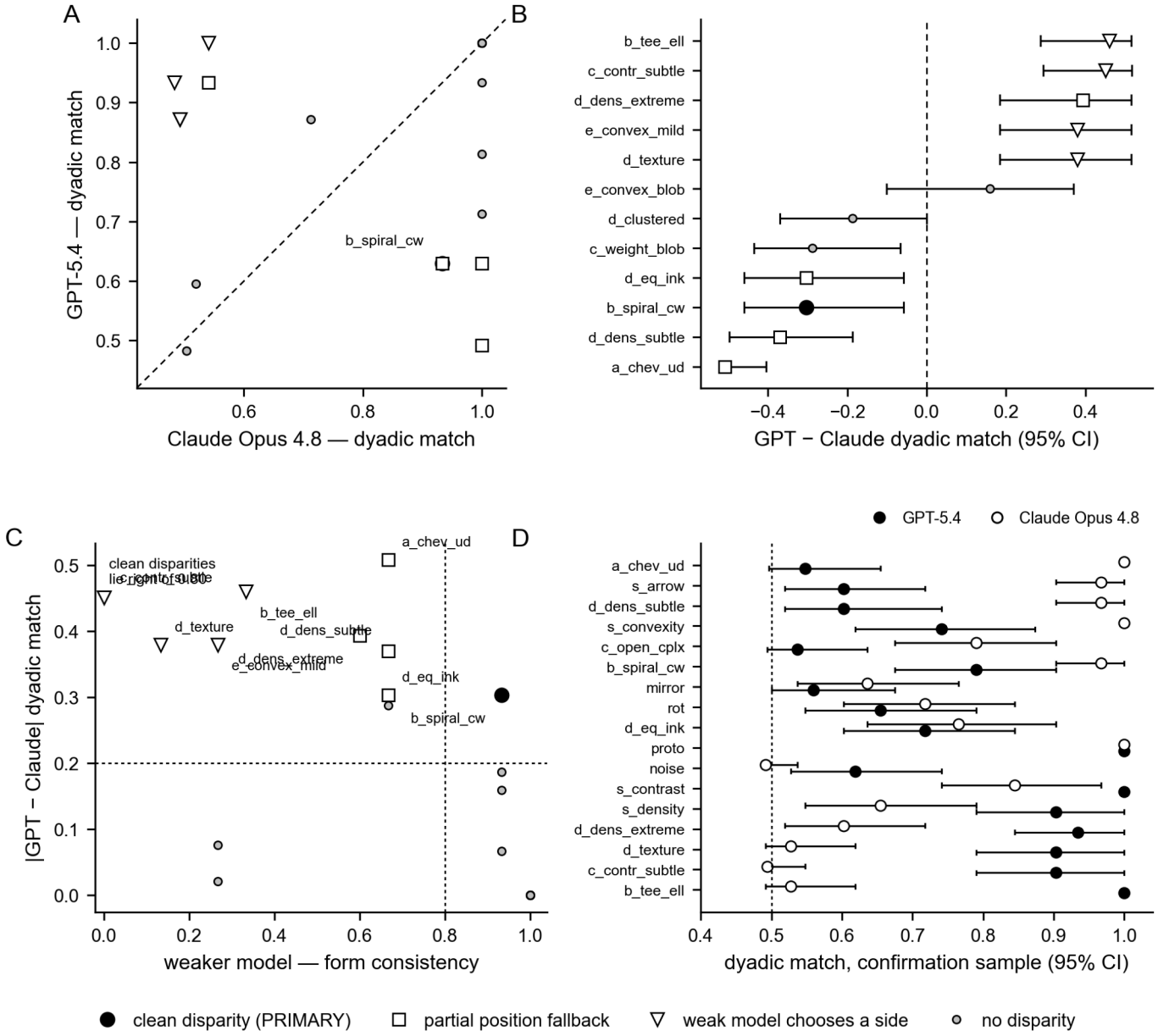
Screen 27 pairs × 2 models × n=30, exact 15/15 AB/BA.
Confirmation selected battery × 2 models × n=60 FRESH trials, exact 30/30 AB/BA, separate allocation seed. Screen and confirmation are never pooled: the screen chose the stimuli, so reusing it would bias them upward.

Quality control

3,660 / 3,660 trials valid. Every model × stimulus cell at its intended N. Exact AB/BA balance in every cell. Zero duplicated calls, zero parse failures, zero API errors.
No trials excluded; no stimulus dropped silently.

Figure 1

(A) GPT vs Claude dyadic match, all 27 screened candidates. (B) Disparity effect sizes, bootstrap 95% CI, 5,000 resamples. (C) The form-consistency diagnostic: clean disparities lie right of 0.80. (D) Confirmation results, fresh n=60 sample.



Form consistency = 1 − | P(choose modal | modal shown LEFT) − P(choose modal | modal shown RIGHT) |. A model choosing by form scores ~1; a model choosing by side scores ~0. Under balanced AB/BA a pure side default reads as ≈0.50 on the identity coding — indistinguishable from indifference on the match rate alone. That is why no low match rate is reported here as indifference unless this diagnostic agrees.

Recommended final battery

All figures from the fresh n=60 confirmation sample. gFC / cFC = form consistency.

GPT > Claude — clean focality disparity

stimulus	GPT	gFC	Claude	cFC	Δ	95% CI	note
d_dens_extreme	0.934	0.93	0.602	0.80	+0.332	[+0.178, +0.452]	clean
s_density	0.903	0.90	0.655	0.97	+0.249	[+0.080, +0.386]	clean
s_contrast	1.000	1.00	0.845	0.97	+0.155	[+0.033, +0.259]	clean, Δ<0.20

Claude > GPT — one clean, the rest differences in kind

stimulus	GPT	gFC	Claude	cFC	Δ	95% CI	note
b_spiral_cw	0.790	0.90	0.967	0.97	-0.176	[-0.304, -0.031]	**clean, Δ<0.20**
a_chev_ud	0.548	0.67	1.000	1.00	-0.452	[-0.503, -0.345]	GPT partial fallback
s_arrow	0.602	0.60	0.967	0.97	-0.364	[-0.463, -0.226]	GPT partial fallback
s_convexity	0.741	0.70	1.000	1.00	-0.259	[-0.381, -0.127]	GPT partial fallback

Diagnostics — large Δ, but the weak model is choosing a side

stimulus	GPT	gFC	Claude	cFC	Δ	95% CI	note
b_tee_ell	1.000	1.00	0.528	0.27	+0.472	[+0.381, +0.508]	NOT a disparity
c_contr_subtle	0.903	0.90	0.494	0.07	+0.410	[+0.294, +0.499]	NOT a disparity
d_texture	0.903	0.97	0.528	0.27	+0.376	[+0.237, +0.475]	NOT a disparity

Controls

stimulus	GPT	gFC	Claude	cFC	Δ	95% CI	note
proto	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	shared-easy positive
rot	0.655	0.97	0.718	0.80	-0.063	[-0.231, +0.115]	intermediate shared
mirror	0.560	0.37	0.636	0.73	-0.076	[-0.217, +0.072]	structured ambiguous
noise	0.619	0.77	0.491	0.13	+0.127	[+0.025, +0.247]	floor

Replication against the earlier screens

stimulus	prior stage	prior Δ	confirmed Δ	replicated
s_arrow	Step 1b	-0.497	-0.364	yes
s_convexity	Step 1b	-0.264	-0.259	yes
c_open_cplx	Step 1b	-0.276	-0.253	yes
s_contrast	Step 1b	+0.370	+0.155	direction only
s_density	Step 1b	+0.186	+0.249	yes
b_spiral_cw	screen	-0.303	-0.176	**direction only**
a_chev_ud	screen	-0.508	-0.452	yes
d_dens_extreme	screen	+0.393	+0.332	yes
d_dens_subtle	screen	-0.370	-0.364	yes
b_tee_ell	screen	+0.460	+0.472	yes
c_contr_subtle	screen	+0.451	+0.410	yes
d_texture	screen	+0.379	+0.376	yes
d_eq_ink	screen	-0.303	-0.048	**direction only**

Screen results and caveats

All 27 targeted candidates, n=30 per model. Sorted by |Δ|.

stimulus	GPT	gFC	Claude	cFC	Δ	95% CI	screen status
a_chev_ud	0.492	0.67	1.000	1.00	-0.508	[-0.517, -0.405]	SECONDARY
b_tee_ell	1.000	1.00	0.540	0.33	+0.460	[+0.287, +0.515]	POSITION_DEFAULT
c_contr_subtle	0.933	0.93	0.483	0.00	+0.451	[+0.294, +0.517]	POSITION_DEFAULT
d_dens_extreme	0.933	0.93	0.540	0.60	+0.393	[+0.184, +0.515]	SECONDARY
d_texture	0.871	1.00	0.492	0.13	+0.379	[+0.184, +0.515]	POSITION_DEFAULT
e_convex_mild	0.871	1.00	0.492	0.27	+0.379	[+0.184, +0.515]	POSITION_DEFAULT
d_dens_subtle	0.630	0.67	1.000	1.00	-0.370	[-0.497, -0.186]	SECONDARY
b_spiral_cw	0.630	0.93	0.933	0.93	-0.303	[-0.460, -0.058]	PRIMARY
d_eq_ink	0.630	0.67	0.933	0.93	-0.303	[-0.460, -0.058]	SECONDARY
c_weight_blob	0.713	0.67	1.000	1.00	-0.287	[-0.434, -0.067]	REJECT
d_clustered	0.814	0.93	1.000	1.00	-0.186	[-0.370, +0.000]	REJECT
e_convex_blob	0.871	0.87	0.713	0.93	+0.159	[-0.101, +0.370]	REJECT
b_s_z	0.595	0.73	0.519	0.27	+0.076	[-0.103, +0.274]	REJECT
c_fill_outline	0.933	0.93	1.000	1.00	-0.067	[-0.186, +0.000]	REJECT
b_hook_mirror	0.483	0.27	0.503	0.20	-0.021	[-0.145, +0.074]	REJECT
a_arrow_lr	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
a_arrow_ud	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
a_arrow_updiag	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
a_tee_rot90	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
a_tee_upright	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
a_tri_ud	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
b_circ_bar	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
b_cross_x	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
c_contr_extreme	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
c_stroke_wt	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
e_closure_c	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
e_open_subtle	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT

Which families were productive

family	n	median |Δ|	both ≥0.95	PRIMARY
A semantic / canonical orientation	7	0.000	6	0
B canonical abstract glyphs	6	0.076	2	1
C contrast / visual weight	5	0.067	2	0
D density / numerosity / texture	5	0.370	0	0
E convexity / closure	4	0.159	2	0

Caveats

— b_spiral_cw is recommended despite a confirmed Δ of 0.176, below the 0.20 target, because it is the only Claude > GPT case where both models track form. Its CI reaches −0.031; a further n=60 would roughly halve that width and settle whether it belongs.

— e_convex_mild (POSITION_DEFAULT, screen Δ +0.379) was cut by the 17-stimulus battery cap and received no confirmation trials. Recorded here rather than dropped silently.

— 12 of 27 candidates sat at 1.000/1.000 for both models. Single-cue manipulations reach ceiling easily; the informative settings are the near-threshold ones.

— Step 1b keyed its conditional-position rates to the layout label, which equals 'modal shown LEFT' only when the modal identity is identity_1. form_consistency was unaffected, so the numbers stay comparable, but the individual conditionals were reversed about half the time. Corrected here.